

STUDENT ACTIVITY

Scenario-Based Use Case Design — Activity

Activity: CNN and Transfer Learning Strategy Design

Q: Read the business narrative below and design three distinct Transfer Learning frameworks. Each framework must explicitly define the Chosen Model, Main Feature, and Common Use:

- One framework optimized for a simple deep architecture.
- One framework optimized for residual learning to reduce the vanishing gradient problem.
- One framework optimized for parallel feature extraction.

Next, using the provided text, describe the fundamental benefits of applying Transfer Learning instead of training a Convolutional Neural Network from scratch in three sentences. Ensure you explicitly mention training time, computational cost, and data requirements.

The Narrative: A healthcare technology company is building an Artificial Intelligence platform for medical imaging and diagnosis. The engineering team has limited training data available for a new, rare disease dataset. To improve classification accuracy and reduce computational cost, the lead architect mandates the reuse of pre-trained Deep Learning models. The team must evaluate VGG16, ResNet50, and InceptionV3 to determine which pre-trained architecture best fits their specific image analysis workflows.

Scenario-Based Use Case Design — Activity

Transfer Learning Frameworks

Framework Optimized for a Simple Deep Architecture

Chosen Model	Main Feature	Common Use
VGG16	Uses a simple and uniform architecture with 16 deep layers and small 3 × 3 convolution filters , making it easy to understand and implement.	Image classification, feature extraction, and medical image analysis where a straightforward deep CNN architecture is preferred.

Framework Optimized for Residual Learning

Chosen Model	Main Feature	Common Use
ResNet50	Uses Residual (Skip) Connections to reduce the vanishing gradient problem , allowing very deep neural networks to train effectively.	Medical image diagnosis, object detection, and other deep learning applications requiring high accuracy and deep architectures.

Framework Optimized for Parallel Feature Extraction

Chosen Model	Main Feature	Common Use
InceptionV3	Uses parallel convolutional filters of different sizes within Inception modules to extract multiple image features simultaneously.	Large-scale image classification, medical imaging, object recognition, and applications requiring efficient feature extraction with lower computational complexity.

Benefits of Transfer Learning

Transfer Learning uses a **pre-trained Convolutional Neural Network (CNN)** instead of training a model from scratch, significantly reducing **training time**. It also lowers **computational cost** because the model reuses previously learned features rather than learning everything again. In addition, Transfer Learning performs well with **limited training data**, making it ideal for applications such as medical imaging where large labeled datasets are often unavailable.

Transfer Learning Frameworks

Framework	Chosen Model	Main Feature	Common Use
Simple Deep Architecture	VGG16	Simple 16-layer architecture using small 3×3 convolution filters .	Image classification and medical image analysis.
Residual Learning	ResNet50	Uses Residual (Skip) Connections to reduce the vanishing gradient problem .	Deep image classification and medical diagnosis.
Parallel Feature Extraction	InceptionV3	Uses parallel convolution filters to extract multiple image features efficiently.	Medical imaging, object recognition, and large-scale image classification.

Benefits of Transfer Learning

Transfer Learning reuses **pre-trained CNN models**, which greatly **reduces training time** compared to training a model from scratch. It **reduces computational cost** by using already learned features instead of learning all features again. It also requires **less training data**, making it highly effective for domains such as medical imaging where labeled data is limited.